

UNIT 1

ANIMAL CLASSIFICATION

Learning objectives:

- Define and classify animals based on their life cycles.
- Identify mammals, reptiles, and other animal groups.
- Explain how animals grow and change over time.
- Describe the life cycle of different animals (e.g. crab).
- Understand the importance of animal adaptations for survival.
- Recognize how animals stay safe from predators.
- Differentiate between animals that lay eggs and those that give birth.
- Explain what a population is and how animals within the same population can look different.
- Describe how animals use their physical features to survive.
- Identify how animals interact within their environment.

KINDS OF ANIMALS

How do we classify animals?

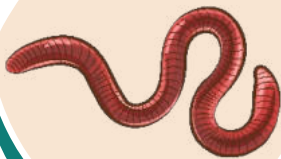
When you classify, you group animals or things that are alike. Scientists classify animals into several groups. In some groups, animals have backbones. In some groups, animals do not have backbones.



Trout



Grasshopper



Worm



Hawk



Touch the back of your neck. Do you feel bumps? They are part of your backbone. Your backbone goes from your hips all the way up to your head.

Squirrel is a mammal



See its backbone

Quick Check

Fill in the blanks.

- Some animals do not have _____.
- A squirrel's backbone goes from its head to its _____.

**The Big Idea: How do animals
grow and change?**



How can we classify animals with backbones?

Scientists classify animals with backbones into smaller groups, such as **mammals, reptiles, amphibians, birds, and fish.**

- **Mammals** are warm-blooded animals that have hair or fur. Examples include humans, cats, tigers, goats, etc.

Mammals like lions give birth to their young.



- **Reptiles** are cold-blooded animals that have scales. Examples include crocodiles, snakes, turtles, lizards, etc.

Most reptiles like alligators lay eggs.



- **Amphibians** are animals that can live both in water and on land. They are cold-blooded and have moist skin. Examples include frogs, toads, and salamanders.

Amphibians like frogs lay eggs in water.



- **Birds** are the only animals with feathers, but not all birds can fly. They are warm-blooded animals that lay eggs. Examples include sparrows, ducks, eagles, parrots, etc.



▲ A bluebird is a bird that can fly.

- **Fish** are cold-blooded animals that live in water. They breathe with body parts called **gills**. Examples include salmon, sharks, goldfish, etc.



▲ Fish have **fins** that help them swim.

Quick Check

- List four animal groups with backbones.

• What group of animals has feathers?



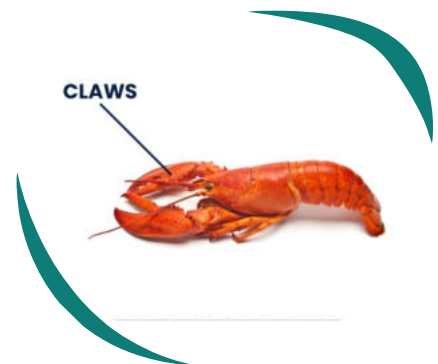
How can we classify animals without backbones?

There are many kinds of animals that do not have backbones. Some of these animals have **shells** or **hard body coverings** that help keep them safe.



◀ The dragonfly has a hard body covering. It uses its wings to fly away from enemies.

The lobster has a hard body covering. It uses its claws to break open food. ▶



◀ The beetle has a hard shell. It has three body parts and six legs.



Some animals without backbones do not have **shells**. They have **soft bodies**. These animals must use other ways to stay safe.



◀ These jellyfish do not have shells. They sting other animals to stay safe.

Quick Check

- Write about two ways that animals without backbones can stay safe.



1. MAMMALS

- Mammals give birth to live young.
- They feed milk to their babies.
- Usually have hair or fur.
- Breathe through lungs and are warm-blooded.
- **Examples:** Humans, cats, dogs, cows, lions, dolphins.



Fun Fact

Whales and dolphins are mammals even though they live in water!

2. REPTILES

- Have dry, scaly skin.
- Most lay eggs.
- Breathe through lungs.
- Cold-blooded.
- **Examples:** Snakes, crocodiles, turtles, lizards.



Fun Fact

Snakes can smell with their tongues!



3. AMPHIBIANS

- Live on both land and water.
- Lay eggs in water.
- Their young ones swim like fish.
- Have moist skin.
- **Examples:** Frogs, toads, newts, salamanders.



Fun Fact

Frogs make a loud, croaky “ribbit!” while toads make a deep, low croak that sounds more like a “grunt.”

4. BIRDS

- Have feathers and wings.
- Most can fly, but not all.
- Lay eggs.
- Warm-blooded and have beaks.
- **Examples:** Pigeons, eagles, parrots, penguins, ostriches.



Fun Fact

Penguins are birds that can't fly but are excellent swimmers!



5. FISH

- Live in water.
- Have scales and fins.
- Lay eggs.
- Breathe through gills.
- Cold-blooded.
- **Examples:** Tuna, sharks, goldfish, trout.

Fun Fact

Fish don't have eyelids, so they can't blink!



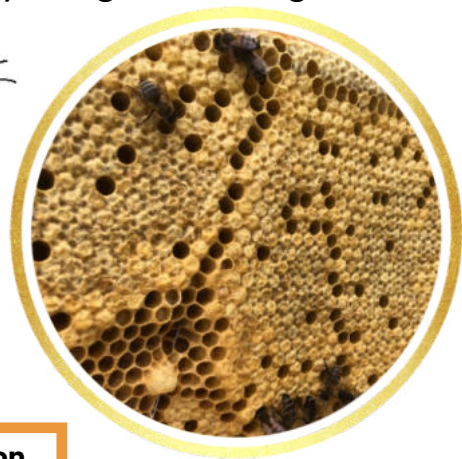
ANIMALS FROM EGGS

Why do animals lay many eggs?

Birds, reptiles, and fish **lay eggs**. Insects and many kinds of sea animals lay eggs, too. Animals lay **many eggs** because some of the young animals will not survive. Many young animals get eaten by other animals.



A queen bee can lay more than 2,000 eggs in one day. ►



Salmon also lay thousands of eggs. They lay their eggs in between rocks in rivers. ►

Salmon



Many animals that **lay eggs** do not care for their young. Female **sea turtles** bury their eggs in the sand and leave.

When the eggs hatch, the **baby turtles** must find their own way to the ocean. Many young turtles get eaten by other animals, such as **seagulls**.



▲ Sometimes people protect baby sea turtles from animals that will eat them.

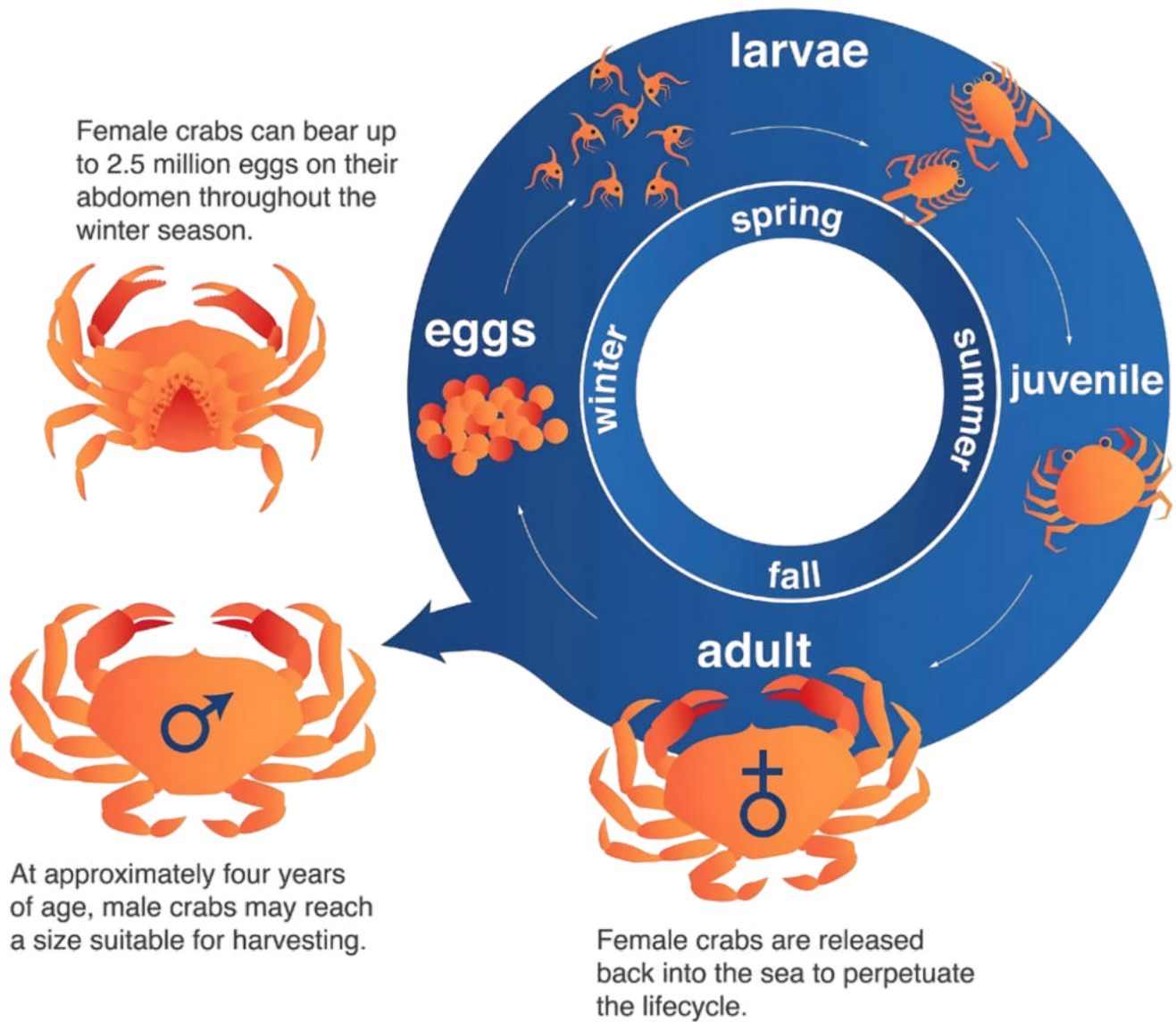
Quick Check

Circle the answers.

- Birds, reptiles, and _____ lay eggs.
 - lions
 - mammals
 - fish
- Sometimes _____ try to help sea turtles.
 - people
 - seagulls
 - mother turtles



CRAB LIFE CYCLE



Quick Check

Fill in the Blanks.

- Most eggs have an outside covering or _____.
- A larva looks _____ from its parents.



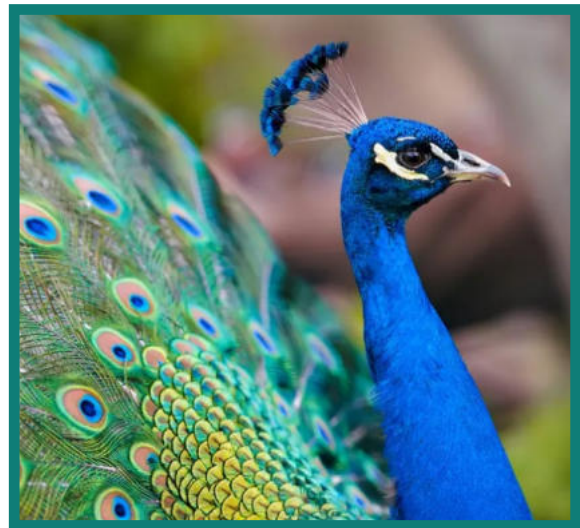
HOW ANIMALS LOOK AND ACT

How do the way animals look and act help them live?

All animals **look and act** in special ways that help them **live**. Sometimes their **colour** or their **body parts** help them. Sometimes the **things they do** can help them.



- ▲ Giraffes have long necks that help them eat leaves in tall trees.



- ▲ Male peacocks have bright feathers and sing loud songs that help females find them.



How do animals stay safe?

Animals do things to **stay safe**, too. Some animals **fly away** when they are in danger. Other animals **blend into their surroundings** or **fight**. The **bites** of some **snakes and spiders** are **poisonous**.

Tortoise



Quick Check

Tell how each animal stays safe.

- Tortoise _____.
- Bird _____.
- Snake _____.



What is a population?

A **population** is a group of the same kind of **animal living near each other**. For example, the **black bears** that live in **Yosemite National Park** are one population. The black bears that live in **Maine** are another population.



▲ Black bears with black fur are found in Maine.



How do populations of animals differ?

Animals in different populations **do not always look the same**. In order to live **safely in different places**, animals can **change in many ways**.



- ▲ Do not be fooled by the name! The black bears that live in Yosemite National Park have brown fur.



- ▲ Some polar bears in Alaska have light fur that helps them hide in the snow.

Quick Check

Write **true** if the sentence is true. Write **false** if the sentence is false.

- Black bears live in different parts of the United States.

- All black bears have black fur. _____



How can we compare animals in the same population?

Animals in the same **population** may **look and act** in different ways. For example, **meerkats** are animals that live in the **African desert**. Some meerkats are **bigger** than others. Some are **more careful**. Others are **more curious**.



How are meerkats alike?

Meerkats are also **alike** in many ways. They **all eat insects** and live **underground in burrows**. They have **long claws** for digging and live in **large groups**.



This meerkat is digging to find insects to eat.

Quick Check

Complete each sentence by writing a fact about meerkats.

- Some meerkats _____.
- All meerkats _____.



Vocabulary Review

Draw a line from each word to its definition.



Classify

1. A group of the same kind of animal living near each other.



Population

2. What a larva becomes when it forms a hard cover around itself?



Larva

3. Cold-blooded animals that have scales.



Pupa

4. To group things that are alike.



Mammals

5. A young animal that hatches from an egg and looks different from its parents.



Reptiles

6. Animals with hair or fur that feed milk to their young ones.



Fill in the blanks.

1. Animals that have hair or fur and feed milk to their young ones are called _____.
2. A group of the same kind of animals living together is called a _____.
3. The hard outer covering that protects some animals is called a _____.
4. A _____ hatches from an egg and looks different from its parents.
5. _____ are cold-blooded animals that have scales.
6. Fish breathe using _____.
7. A _____ bee can lay more than 2,000 eggs in one day.
8. Meerkats dig burrows to _____.



Critical Thinking

Why do you think vertebrates generally grow larger and live longer than invertebrates? How does having a backbone provide advantages to animals?



Write “T” for True or “F” for False.

1. Mammals lay eggs.
2. All animals have the same life cycle.
3. Some animals change colour to hide from predators.
4. Fish breathe through their lungs.
5. A larva looks exactly like its parents.
6. Some black bears have brown fur instead of black.
7. A tortoise can pull its head inside its shell for protection.
8. Giraffes have long necks to help them eat from tall trees.
9. A mother sea turtle takes care of her babies after they hatch.



Choose the correct answer.

1. What is the outer covering of reptiles called?

- Fur
 - Scales
 - Skin
-

2. Which of these animals lay eggs?

- Birds
 - Cows
 - Cats
-

3. What is the purpose of gills in fish?

- To eat food
 - To breath oxygen from water
 - To swim faster
-

4. A larva turns into a _____ before becoming an adult.

- Pupa
 - Egg
 - Tadpole
-





5. What do giraffes eat?

- Leaves of tall trees
- Fish
- Fruits

6. What is the special feature of a giraffe?

- Long tail
- Long neck
- Long ears

7. Which of these is a mammal?

- Frog
- Human
- Fish

8. How do birds stay safe?

- Running fast
 - Flying away
 - Hiding in Holes
-





9. What do meerkats eat?

- Insects
- Fruit
- Fish

10. Why do animals lay many eggs?

- To decorate their nest
- To increase the chances of survival
- To keep warm



Answer the following questions.

Q1. What are mammals? Give two examples.

Q2. How do reptiles differ from mammals?

Q3. Why do sea turtles lay many eggs?

Q4. How do fish breathe?

Q5. What is a population?

Q6. What is the life cycle of a crab?

DO YOU KNOW?

- Vertebrates are animals that have a **backbone (spine)**.

Examples: **Lions, Fish, Birds, Frogs, Snakes.**

- **Invertebrates** are animals that **do not have a backbone.**

Examples: **Worms, Insects, Jellyfish, Octopuses, Crabs.**



Can You Guess?

- Is a **spider** vertebrate or invertebrate?
- Does a **snake** have a backbone?

VERTEBRATES



VERTEBRATE

INVERTEBRATE



VERTEBRATE

INVERTEBRATE



VERTEBRATE

INVERTEBRATE



VERTEBRATE

INVERTEBRATE



VERTEBRATE

INVERTEBRATE

INVERTEBRATES



VERTEBRATE

INVERTEBRATE



VERTEBRATE

INVERTEBRATE



VERTEBRATE

INVERTEBRATE



VERTEBRATE

INVERTEBRATE



VERTEBRATE

INVERTEBRATE



1. Drag and drop in correct place.

Dog **Crocodile** **Spider** **Worm** **Octopus** **Shark**



2. Write the name of the animal in the correct place.

VERTEBRATES	INVERTEBRATES



**Observe, Identify, and Classify Animals
in Your Environment**



ACTIVITY

- Go to your garden, school playground, or a nearby park and observe 5 different animals or insects. Write down their names, describe their physical features, and classify them as vertebrates or invertebrates. Also identify whether it is a mammal, bird, reptile, insect, amphibian, or fish.

